Aerial Object Classification and Detection

Capstone Project Report

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1. Introduction

With the rapid advancement in aerial technologies such as drones and surveillance systems, the need for intelligent monitoring systems has become increasingly important. In many real-world environments, it is essential to distinguish between natural flying objects like birds and artificial aerial vehicles such as drones.

This project focuses on building a deep learning-based system capable of classifying aerial images into two categories: **Bird** and **Drone**. By leveraging computer vision and transfer learning techniques, the system aims to provide accurate and efficient classification suitable for real-world applications.

The project demonstrates how deep learning models can be applied to solve practical problems in domains such as aviation safety, defense surveillance, and environmental monitoring.

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2. Problem Statement

In critical environments such as airports, military zones, and wildlife monitoring areas, distinguishing between birds and drones is a significant challenge.

Incorrect classification can lead to:

- Failure to detect unauthorized drones, posing security threats
- False alarms caused by birds, leading to unnecessary interventions
- Increased risk of bird strikes in aviation

The objective of this project is to develop a reliable and accurate classification system that can differentiate between birds and drones using aerial images.

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3. Objectives

The key objectives of this project are:

- To build a binary image classification model for distinguishing between birds and drones
- To implement both a custom Convolutional Neural Network (CNN) and a transfer learning model
- To evaluate and compare model performance using standard metrics
- To achieve high accuracy and generalization on unseen data
- To deploy the trained model using a Streamlit-based interactive dashboard
- To demonstrate real-time prediction capability

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4. Dataset Description

The dataset used in this project consists of labeled RGB images categorized into two classes: **Bird** and **Drone**. The dataset is divided into three subsets:

Dataset Split	Number of Images
Training	2662
Validation	442
Testing	215

The dataset is relatively balanced, with a comparable number of images for both classes. This helps in reducing bias during model training and improves classification performance.

Each image is stored in a structured directory format:

- train/bird
- train/drone
- valid/bird
- valid/drone
- test/bird

- test/drone

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5. Data Preprocessing and Augmentation

Before feeding the images into the model, several preprocessing steps were applied:

- **Resizing:** All images were resized to 224 × 224 pixels
- **Normalization:** Pixel values were scaled to the range [0, 1]
- **Label Encoding:** Binary labels were assigned (0 for Bird, 1 for Drone)

To improve model generalization, light data augmentation techniques were applied:

- Horizontal flipping
- Small rotations
- Slight zoom

Since the dataset already contained similar and repeated images, only minimal augmentation was used to avoid overfitting.

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6. Model Development

Two models were developed and compared:

6.1 Custom CNN

A Convolutional Neural Network was built from scratch with:

- Multiple convolutional layers
- Max pooling layers
- Fully connected layers
- Dropout for regularization

This model serves as a baseline for comparison.

6.2 Transfer Learning (MobileNetV2)

A transfer learning approach was implemented using **MobileNetV2**, a pre-trained model on ImageNet.

Key features:

- Efficient architecture
- Strong feature extraction capability
- Faster convergence

The base model was frozen initially, and a custom classification head was added.

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7. Training Configuration

The models were trained using the following configuration:

- **Optimizer:** Adam
- **Loss Function:** Binary Crossentropy
- **Batch Size:** 32
- **Epochs:** 15

Regularization techniques:

- EarlyStopping
- ReduceLROnPlateau
- Dropout

These techniques helped prevent overfitting and improve model stability.

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8. Model Evaluation

The models were evaluated using the following metrics:

- Accuracy
- Precision
- Recall
- F1-score

Performance Comparison

Metric	Custom CNN	MobileNetV2
Accuracy	91–93%	96.6%
Precision	0.91	0.96
Recall	0.92	0.96
F1-score	0.91	0.96

The transfer learning model significantly outperformed the custom CNN.

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9. Results and Discussion

The MobileNetV2 model achieved high accuracy and demonstrated strong generalization on unseen data.

Observations:

- Better feature extraction compared to custom CNN
- Faster convergence during training
- Reduced overfitting

Some misclassifications occurred in cases where:

- drones resembled birds at a distance
- images had low resolution

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10. Deployment (Streamlit Dashboard)

The final model was deployed using **Streamlit**, providing an interactive user interface.

Features of the dashboard:

- Project overview
- Dataset statistics and visualization
- Model performance metrics
- Live image upload and prediction
- Confidence score display

This allows real-time interaction and demonstration of the model.

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11. Applications

The system can be applied in several real-world scenarios:

- Airport bird-strike prevention
- Defense and surveillance systems
- Wildlife monitoring
- Environmental research
- Airspace safety monitoring

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12. Limitations

- Limited dataset diversity
- Similar-looking objects may cause confusion
- No real-time video processing
- No object detection (bounding boxes) implemented

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13. Future Work

Future improvements may include:

- Integration of YOLOv8 for object detection
- Expansion of dataset with more variations
- Real-time video processing system
- Deployment on edge devices
- Multi-class classification for more aerial objects

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14. Conclusion

This project successfully demonstrates the use of deep learning for aerial image classification.

The transfer learning approach using MobileNetV2 proved to be highly effective, achieving strong accuracy and generalization.

The developed system can be extended and deployed in real-world safety and monitoring applications, making it a valuable solution in modern computer vision systems.